

Kragujevac, Serbia

WMO: 132780

Lat: 44.033N

Lon: 20.933E

Elev: 607

StdP: 14.38

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	17.1	8.9	8.9	12.8	13.5	11.2	17.4	19.3	44.6	16.0	39.8	2.0	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.5	95.0	70.7	91.7	70.6	88.4	69.9	73.8	88.6	72.4	86.9	71.0	85.1	4.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.3	110.0	80.8	67.7	104.0	79.1	66.2	98.6	77.0	37.7	88.5	36.4	86.6	35.1	85.3	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	11.5	9.7	3.0	98.2	7.8	3.4	-2.6	100.7	-7.2	102.6	-11.6	104.5	-17.3	107.0
			2.4	76.8	7.7	1.7	-3.2	78.0	-7.7	79.1	-12.0	80.0	-17.6	81.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.8	34.3	38.2	46.3	55.3	62.1	69.8	73.0	73.4	64.8	54.5	47.2	37.5	(6)
	DBStd	15.46	9.70	9.88	8.14	6.36	6.28	6.00	5.37	5.30	7.28	7.62	8.52	8.52	(7)
	HDD50	1611	489	343	167	22	1	0	0	0	1	43	149	396	(8)
	HDD65	4553	952	752	580	297	132	23	5	4	90	330	535	853	(9)
	CDD50	3354	2	11	52	181	375	593	713	724	446	184	65	8	(10)
	CDD65	821	0	0	0	6	41	166	252	264	86	5	1	0	(11)
	CDH74	9107	0	0	3	130	477	1766	2794	2933	907	95	2	0	(12)
	CDH80	3750	0	0	0	23	92	662	1249	1388	327	9	0	0	(13)
(14) Wind	WSAvg	3.6	3.9	4.0	4.6	3.8	3.3	3.3	3.2	3.1	3.1	3.1	3.7	4.0	(14)
(15) Precipitation	PrecAvg	25.7	1.7	1.5	1.9	2.2	2.6	2.9	2.6	2.1	2.4	2.2	1.9	2.0	(15)
	PrecMax	39.8	3.7	3.1	4.8	5.6	9.1	6.0	10.7	5.2	5.8	4.7	4.2	5.0	(16)
	PrecMin	15.4	0.4	0.4	0.1	0.2	0.8	1.0	0.4	0.0	0.5	0.0	0.1	0.2	(17)
	PrecStd	5.6	0.9	0.7	1.3	1.2	1.7	1.3	2.0	1.4	1.3	1.3	1.0	1.2	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.3	67.9	73.1	83.2	86.5	93.9	99.1	99.7	93.8	80.5	71.9	63.2	(19)
		MCWB	50.8	53.6	56.5	63.6	67.0	72.5	71.0	71.8	69.7	63.6	59.4	54.2	(20)
	2%	DB	54.9	61.6	69.0	78.2	82.4	90.3	94.7	95.4	88.5	76.7	68.0	58.3	(21)
		MCWB	47.8	51.2	53.8	61.6	66.3	72.2	70.8	70.6	67.5	62.2	57.1	50.8	(22)
	5%	DB	51.4	56.9	65.0	73.0	79.4	87.2	91.2	91.8	83.0	73.0	64.3	54.0	(23)
		MCWB	45.8	48.6	52.3	58.1	65.1	70.6	70.8	70.0	66.0	60.7	55.2	48.1	(24)
	10%	DB	47.6	52.7	60.3	69.1	76.0	83.7	87.5	87.9	79.1	68.5	60.6	49.6	(25)
		MCWB	43.4	46.2	49.7	56.2	63.6	69.3	70.1	69.3	64.3	58.6	53.3	45.0	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	54.3	57.5	65.1	69.6	76.0	75.2	75.3	71.1	65.7	60.1	54.8	(27)
		MCDB	58.4	65.7	71.6	80.5	81.9	89.1	89.5	91.0	89.8	78.3	70.2	61.9	(28)
	2%	WB	48.5	51.5	54.7	61.8	67.6	73.6	73.6	73.2	68.6	63.1	57.6	51.3	(29)
		MCDB	54.3	60.7	66.2	75.7	79.9	86.7	88.6	89.1	84.7	74.5	66.9	57.6	(30)
	5%	WB	46.1	49.0	52.6	59.4	65.9	71.8	72.1	71.7	67.0	61.1	55.8	48.1	(31)
		MCDB	50.8	56.0	63.3	71.1	77.8	84.7	86.3	87.5	80.7	71.2	63.2	53.3	(32)
	10%	WB	43.6	46.5	50.6	57.2	64.2	70.0	70.7	70.2	65.3	59.3	53.8	45.2	(33)
		MCDB	47.5	52.4	59.8	68.0	74.9	81.9	84.8	85.3	76.8	67.4	59.9	49.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.6	14.5	17.6	20.2	20.1	20.4	22.5	24.4	21.0	18.3	16.1	11.9	(35)
		MCDBR	17.4	24.0	26.7	29.5	27.0	26.6	30.5	32.4	28.7	26.5	21.6	18.8	(36)
	5% WB	MCWBR	11.8	14.6	14.3	15.0	12.9	11.1	11.2	11.6	11.6	13.8	12.2	12.5	(37)
		MCWBR	16.9	21.9	24.0	26.1	24.5	24.3	24.7	26.9	25.0	24.2	19.9	17.3	(38)
(40) Clear-Sky Solar Irradiance	taub		0.340	0.366	0.402	0.460	0.450	0.443	0.444	0.457	0.434	0.394	0.356	0.336	(40)
		taud	2.332	2.235	2.201	2.059	2.149	2.211	2.208	2.170	2.204	2.308	2.364	2.357	(41)
	Ebn at Noon		243	256	262	255	262	264	262	253	249	243	235	232	(42)
		Edh at Noon	26	34	40	50	47	44	44	44	40	31	25	23	(43)
(44) All-Sky Solar Radiation	RadAvg		420	681	1052	1407	1711	1925	1970	1766	1237	851	529	363	(44)
	RadStd		61	87	131	160	174	190	154	168	127	107	56	54	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (47 neighbors)	+1.21	N/A	N/A	N/A	+0.70	N/A	N/A	-281	+189	N/A

Nomenclature: See separate page